In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("dataset/train.csv")

print("Dataset Shape:", df.shape)
print("\nFirst Five Rows:")
display(df.head())

print("\nDataset Information:")
df.info()

print("\nStatistical Summary:")
display(df.describe())

print("\nMissing Values:")
display(df.isnull().sum())
```

```
Dataset Shape: (891, 12)

First Five Rows:
```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>PassengerId</th> <th>Survived</th> <th>Pclass</th> <th>Name</th> <th>Sex</th> <th>Age</th> <th>SibSp</th> <th>Parch</th> <th>Ticket</th> <th>Fare</th> <th>Cabin</th> <th>Embarked</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 3 | Braund, Mr. Owen Harris | male | 22.0 | 1 | 0 | A/5 21171 | 7.2500 | NaN | S |
| 2 | 1 | 1 | Cumings, Mrs. John Bradley (Florence Briggs Th... | female | 38.0 | 1 | 0 | PC 17599 | 71.2833 | C85 | C |
| 3 | 1 | 3 | Heikkinen, Miss. Laina | female | 26.0 | 0 | 0 | STON/O2. 3101282 | 7.9250 | NaN | S |
| 4 | 1 | 1 | Futrelle, Mrs. Jacques Heath (Lily May Peel) | female | 35.0 | 1 | 0 | 113803 | 53.1000 | C123 | S |
| 5 | 0 | 3 | Allen, Mr. William Henry | male | 35.0 | 0 | 0 | 373450 | 8.0500 | NaN | S |

```
Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    str
 4   Sex          891 non-null    str
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    str
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str
 11  Embarked     889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB

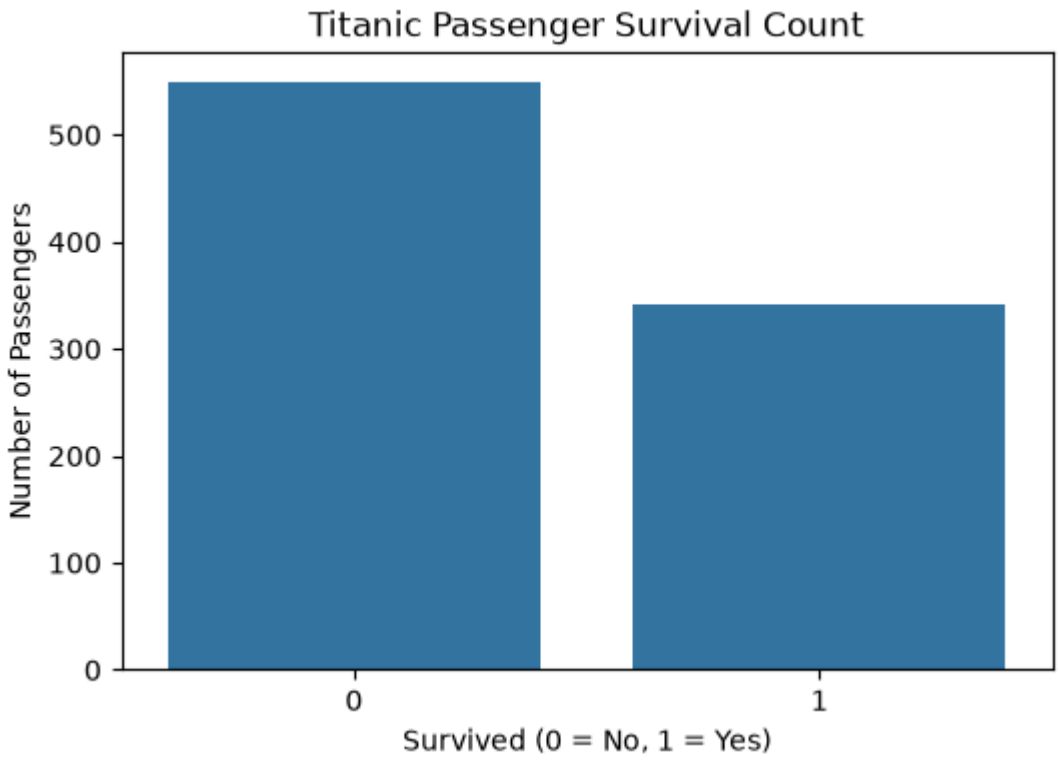
Statistical Summary:
```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>PassengerId</th> <th>Survived</th> <th>Pclass</th> <th>Age</th> <th>SibSp</th> <th>Parch</th> <th>Fare</th> </thead> <tbody> <th>count</th> <th>mean</th> <th>std</th> <th>min</th> <th>25%</th> <th>50%</th> <th>75% </th> <th>max</th> </tbody> <tbody></tbody>

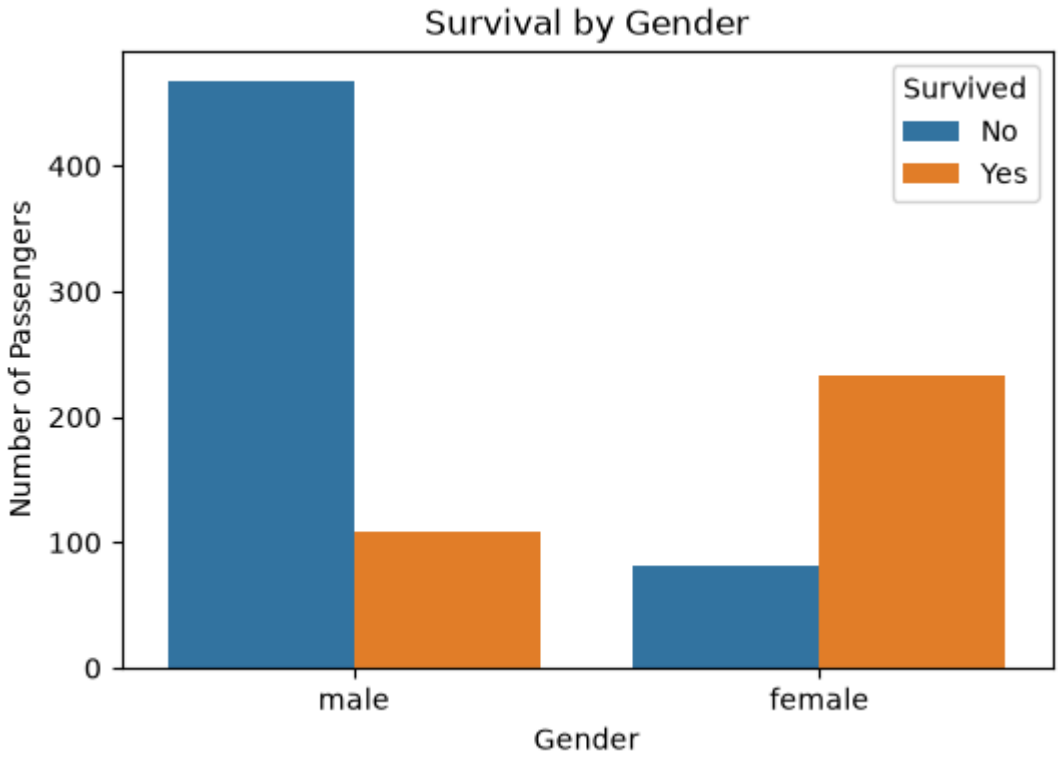
| | | | | | | |
|---|---|---|---|---|---|---|
| 891.000000 | 891.000000 | 891.000000 | 714.000000 | 891.000000 | 891.000000 | 891.000000 |
| 446.000000 | 0.383838 | 2.308642 | 29.699118 | 0.523008 | 0.381594 | 32.204208 |
| 257.353842 | 0.486592 | 0.836071 | 14.526497 | 1.102743 | 0.806057 | 49.693429 |
| 1.000000 | 0.000000 | 1.000000 | 0.420000 | 0.000000 | 0.000000 | 0.000000 |
| 223.500000 | 0.000000 | 2.000000 | 20.125000 | 0.000000 | 0.000000 | 7.910400 |
| 446.000000 | 0.000000 | 3.000000 | 28.000000 | 0.000000 | 0.000000 | 14.454200 |
| 668.500000 | 1.000000 | 3.000000 | 38.000000 | 1.000000 | 0.000000 | 31.000000 |
| 891.000000 | 1.000000 | 3.000000 | 80.000000 | 8.000000 | 6.000000 | 512.329200 |

```
Missing Values:
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

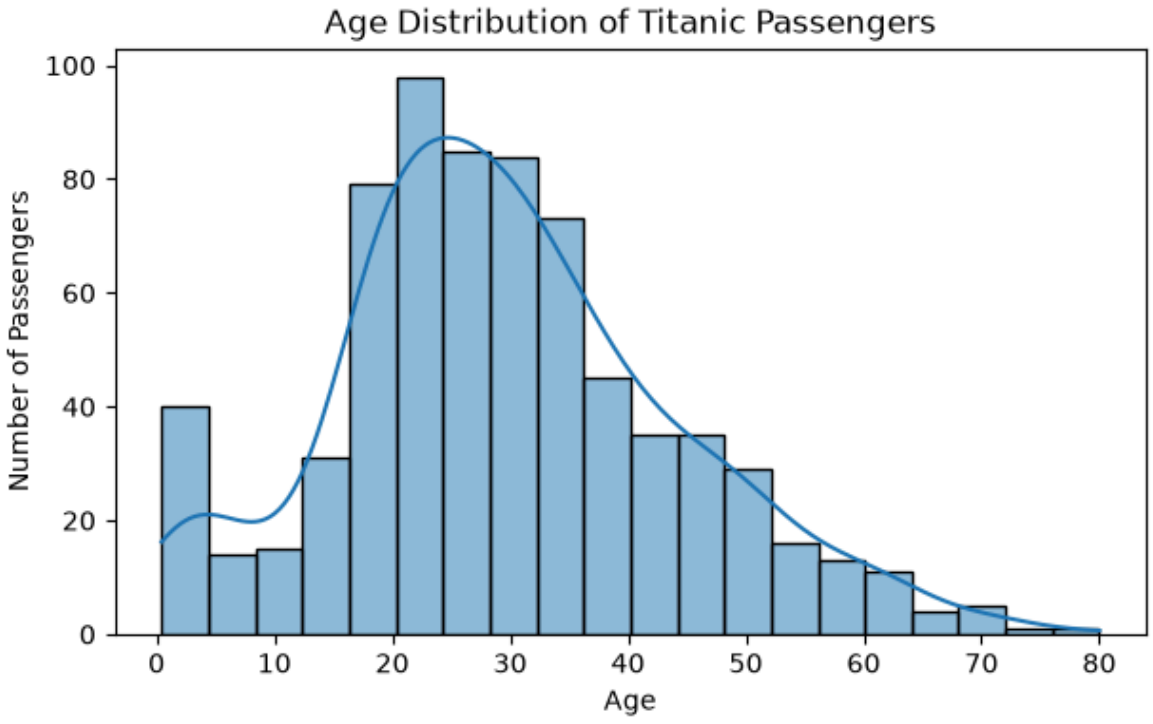
In [2]:
```
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="Survived")
plt.title("Titanic Passenger Survival Count")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Number of Passengers")
plt.show()
```



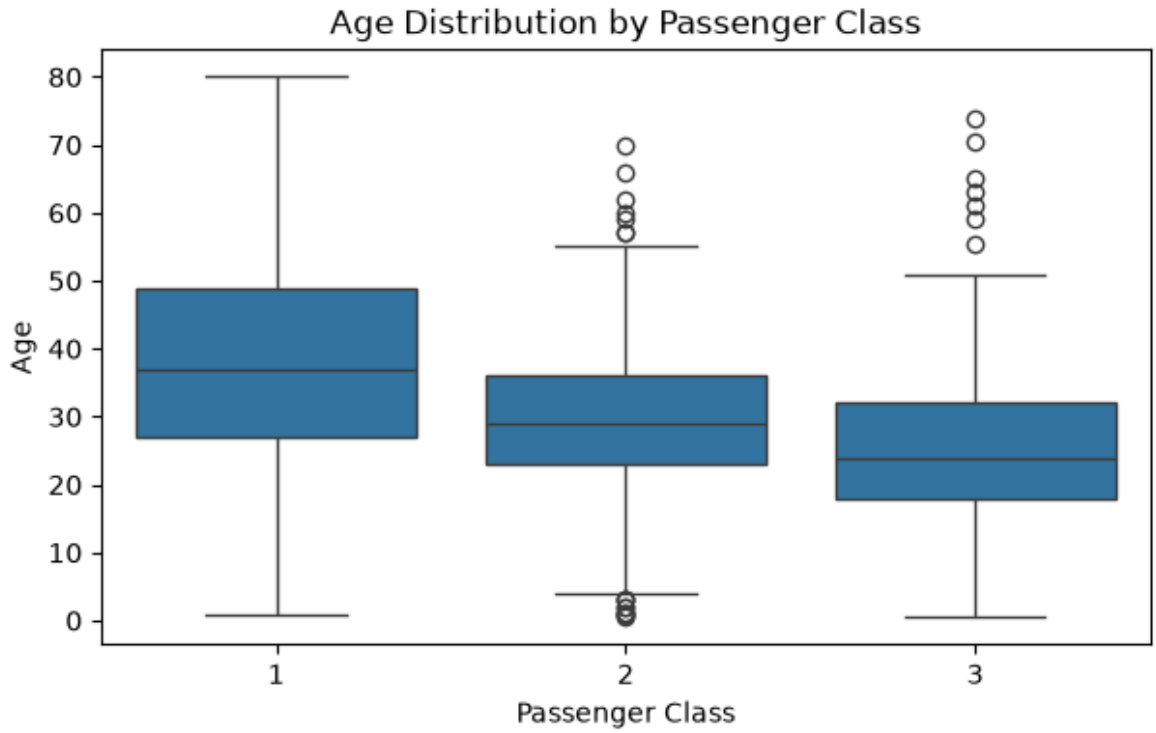

In [3]:
```
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="Sex", hue="Survived")
plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.legend(title="Survived", labels=["No", "Yes"])
plt.show()
```




In [4]:
```
plt.figure(figsize=(7, 4))
sns.histplot(data=df, x="Age", bins=20, kde=True)
plt.title("Age Distribution of Titanic Passengers")
plt.xlabel("Age")
plt.ylabel("Number of Passengers")
plt.show()
```




In [5]:
```
plt.figure(figsize=(7, 4))
sns.boxplot(data=df, x="Pclass", y="Age")
plt.title("Age Distribution by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Age")
plt.show()
```

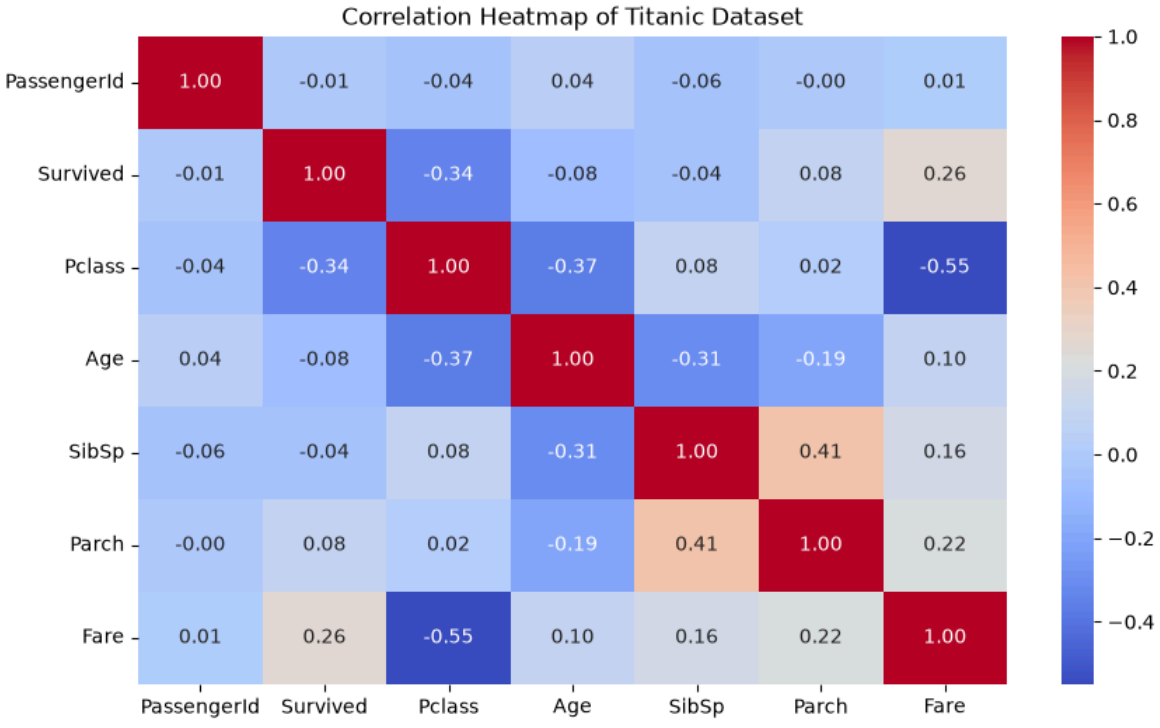



In [6]:
```
plt.figure(figsize=(10, 6))

numeric_df = df.select_dtypes(include="number")

sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Heatmap of Titanic Dataset")
plt.show()
```



<h2 id="EDA-Findings">EDA Findings</h2>

1. More passengers died than survived.
2. Female passengers had a higher survival rate than male passengers.
3. The dataset contains passengers of different age groups.
4. Passenger ages varied across the three passenger classes.
5. The correlation heatmap shows relationships between numerical variables.

<h2 id="Conclusion">Conclusion</h2> Exploratory Data Analysis of the Titanic dataset helped identify patterns in passenger survival, gender, age, and passenger class. Graphs and statistical summaries made the dataset easier to understand.

In [ ]: